TSM 098

Homework 8

Percents and equations in context

Topics 35-37 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Work without a calculator before checking. Give exact, simplified answers.

Part, whole, and percent

1. Convert: (a) 35% to a decimal; (b) 0.08 to a percent; (c) 3/4 to a percent; (d) 125% to a decimal.

2. Find 18% of 150. Write the part = rate × whole equation.

3. 42 is what percent of 120? Identify the whole.

4. 36 is 45% of what number? Define the unknown and solve an equation.

Changes and successive percents

5. A $60 item is discounted by 25%. Find the discount and sale price.

6. A $50 item is taxed at 8%. Find the tax and total.

7. Enrollment grows from 80 to 100. Find the percent increase and explain why you divide by 80.

8. A price falls from $120 to $90. Find the percent decrease. What percent increase takes $90 back to $120?

9. A $100 item is discounted 20%, then taxed 5% on the discounted price. Find the total. Explain why subtracting 15% from the original price gives a different result.

Translate, solve, and check

10. A rectangle has perimeter 34 cm. Its length is 3 cm more than its width. Define a variable, write an equation, and find both dimensions.

11. A club sells 20 tickets: student tickets cost $4 and general tickets cost $7. Total revenue is $107. Define variables, write a system, solve, and check both the ticket count and revenue.

12. A printer rental costs $12 plus $0.05 per page. The total is $22. Write and solve an equation for the number of pages. Check the units and total cost.